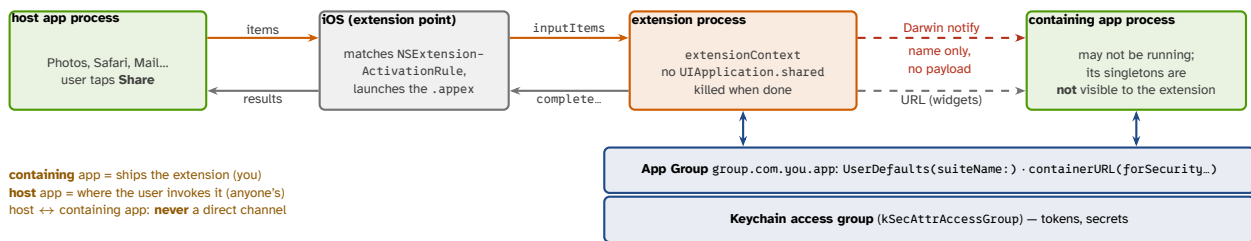


In one line: An extension is a .appex inside the **containing app**, launched by the system for a **host app in its own process** — own sandbox, bundle id, entitlements, a far smaller memory limit. It shares **no memory** with its app: they meet only in an **App Group**, a **Keychain access group**, or through the system.



Extension points (common)

Share / Action	share sheet in any app; Action can hand back modified items
Widget	WidgetKit timeline, archived views, no live process
Notif. Service	mutable-content: 1 push, ≈30 s to edit/decrypt/attach, no UI
Notif. Content	custom VC for an expanded notification, per category
Keyboard	not used for secure/phone-pad fields; no network or App Group without <i>Full Access</i>
App Intents / Intents	Siri, Shortcuts, Spotlight actions
File Provider	a cloud drive inside the Files app
Network Ext.	VPN packet tunnel, content filter, DNS proxy; special entitlement
also	Safari web ext., Photo Editing, iMessage, Call Directory, Broadcast Upload

How it works

- A **separate target**: bundle id under the app's (com.you.app.share), its own profile and entitlements. Info.plist NSExtension → NSExtensionPointIdentifier, NSExtensionPrincipalClass or ..MainStoryboard, NSExtensionAttributes.
- **NSExtensionContext**: inputItems → NSExtensionItem. attachments → NSItemProvider (hasItemConforming..., async loadItem); finish with completeRequest(returningItems:) or cancelRequest(withError:).
- **Lifetime** is the system's: launched on demand, killed soon after completing. **API limits**: UIApplication.shared is marked unavailable; a shared framework sets **"Allow app extension API only"** so misuse fails at build time. extensionContext.open(_:) is honoured only by some points — a share extension cannot open its app; widgets use widgetURL/Link.
- **Memory** ceilings: per point, undocumented, far below an app's — widgets and NSEs are famously tight. Downsample, don't cache.
- **Shared data**: App Group entitlement on *both* targets; files or a SQLite/Core Data store in containerURL, writes through NSFFileCoordinator; the app nudges widgets with WidgetCenter.shared.reloadTimelines(ofKind:).
- **Long upload**: background URLSession + sharedContainerIdentifier = the group; events go to the containing app (handleEventsForBackgroundURLSession).
- **Debug**: run the extension scheme, pick a host app (Xcode attaches when invoked) or *Attach to Process by Name*; log with Logger + Console.app; the memory gauge shows the extension's own limit.

Example — share extension → its app

```
final class ShareVC: UIViewController {
    override func viewDidLoad() { super.viewDidLoad(); Task {
        let ctx = extensionContext!, id = UTType.url.identifier
        let item = ctx.inputItems.first as? NSExtensionItem
        guard let p = item?.attachments?.first(where: {
            $0.hasItemConformingToTypeIdentifier(id) }),
            let url = try? await p.loadItem(
                forTypeIdentifier: id, options: nil) as? URL
        else { return ctx.cancelRequest(
            withError: CocoaError(.fileReadUnknown)) }
        let shared = UserDefaults(suiteName: "group.com.you.app")
        shared?.set(url.absoluteString, forKey: "pendingURL")
        ctx.completeRequest(returningItems: nil) // may die now
    } } }
```

Traps

- ! UserDefaults.standard in app + widget → two different files. Use the suite, entitlement on *both* targets.
- ! Singletons and caches are per process: after the extension writes, the app's cache is stale — reload on foreground or on a Darwin notification.
- ! URLSession.shared then completeRequest — the process is killed mid-request. Use a background session.
- ! Shared SQLite: a suspended process holding its file lock is killed (0xdead10cc); coordinate writers.
- ! NSExtensionActivationRule = TRUEPREDICATE is for development only — App Review rejects it.

Remember

Own process · own sandbox · shared disk, never shared memory.

Which channel for what

app → widget	write the App Group store, reloadTimelines(ofKind:)
login token	Keychain access group (an App Group id works too)
big upload	background session, sharedContainerIdentifier
"I wrote"	Darwin notification (name only) → read the group store
open the app	widgets: widgetURL / Link; share ext: no

NSExtensionPointIdentifier

Share · Action	com.apple.share-services · com.apple.ui-services
Widget	com.apple.widgetkit-extension
Notif. Service	com.apple.usernotifications.service
Notif. Content	com.apple.usernotifications.content-extension
Keyboard	com.apple.keyboard-service
File Provider	com.apple.fileprovider-nonui
Photo Editing	com.apple.photo-editing

Related: widgets-live-activities (timelines, budget) · push-notifications (Notification Service Extension) · keychain-secure-enclave (access groups) · background-execution (0xdead10cc, background URLSession) · app-intents · deep-links-universal-links